TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Single Yarn Strength, Lea Strength and C.S.P., Irregularity and Yarn Fault Analysis

The yarn itself on trial: how strongly a single end and a full lea break, how evenly the mass is distributed along the length, and how its rare faults are classified for winding and clearing.

TE-3104 · QUALITY EVALUATION OF FIBRE AND YARN SESSIONAL · EXPERIMENT 5 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to determine (a) the single-end breaking load and tenacity of a yarn and its lea (skein) breaking load, combined into the count–strength product (C.S.P.), (b) the mass irregularity of the yarn — U% and CV_m — with its imperfections (thin places, thick places and neps) on an evenness tester, and (c) the classification of yarn faults by size and length in the outline of the Uster Classimat system.

1.1 Single-end strength and tenacity

A short gauge length of yarn (500 mm is standard) is extended at a constant rate until it breaks; the instrument records breaking load and elongation. The raw load depends on how thick the yarn is, so results are compared as **tenacity** — breaking load per unit linear density:

$$\text{Tenacity (cN/tex)} = \text{breaking load (cN)} / \text{linear density (tex)}, \quad \text{tex} = 590.5 / \text{Ne}$$

Because yarn breaks at its weakest tested place, a mean of many breaks (at least 20, and preferably 50) with its coefficient of variation is reported: the CV of strength predicts ends-down in weaving almost as directly as the mean predicts fabric strength.

1.2 Lea strength and C.S.P.

The traditional mill test breaks a whole lea — the same 120-yard lea wound for the count test in Experiment 2 — looped over two hooks of a pendulum or motorised lea tester. A lea break averages the strength of 80 parallel strands, so it is steadier than any single-end figure, though a little lower per strand than the mean single-end load would suggest, because the strands share the load unevenly and

the weakest strands fail first and shed load onto the rest. Multiplying the lea load by the count gives the **count–strength product**:

$$\text{C.S.P.} = \text{lea breaking load (lb)} \times \text{yarn count (Ne)}$$

C.S.P. is a deliberately count-compensated quality number: finer yarns inevitably break at lower loads, and the multiplication asks whether the yarn is strong *for its count*. For carded cotton yarns a C.S.P. around 2,000 is fair, 2,200–2,500 good, and combed yarns run higher.

1.3 Irregularity: U% and CV_m

No spun yarn has constant mass along its length; the drafting of a random fibre assembly guarantees a baseline of variation, and machinery adds more. The evenness tester (Uster principle) draws the yarn through a capacitive measuring field: the capacitance follows the mass present, producing a continuous mass signal from which the instrument computes:

- **U%** — the mean absolute deviation of mass from the mean, as a percentage (the "unevenness").
- **CV_m%** — the standard deviation of mass as a percentage of the mean. For a roughly normal mass distribution the two are linked by $CV_m \approx 1.25 \times U\%$.
- **Imperfections (IPI)** — counts per kilometre of places crossing set thresholds: thin places at −50% of mean mass, thick places at +50%, and neps at +200%. These are the frequent, small faults of spinning, reported per km so yarns of any test length compare fairly.

The mass diagram (the drawn trace of the signal) and its spectrogram separate random variation from periodic faults: a wave of fixed wavelength in the spectrogram points to one rotating element — a damaged roller or gear — whose circumference or gearing sets that wavelength.

1.4 Fault classification: Classimat in outline

Imperfections are frequent and small; **faults** are rare and objectionable — the long thick places, slubs and foreign matter that survive to spoil fabric or break ends in winding. Yarn clearers on the winder scan every metre and cut out faults above a clearing limit. The Classimat system classifies removed (and reference) faults on a grid of **length against cross-sectional increase**: short faults are grouped in classes A–D of increasing size (from small, short thick places up to very large short faults), long faults in length classes commonly labelled E, F and G, and long thin faults as H and I. In the laboratory, a fault board or the clearer's own classing data is tallied into this grid and reported as faults per 100 km, so a spinning mill can tell whether its problem is many small faults (fibre and preparation) or a few long ones (piecing, roller laps and machine defects).

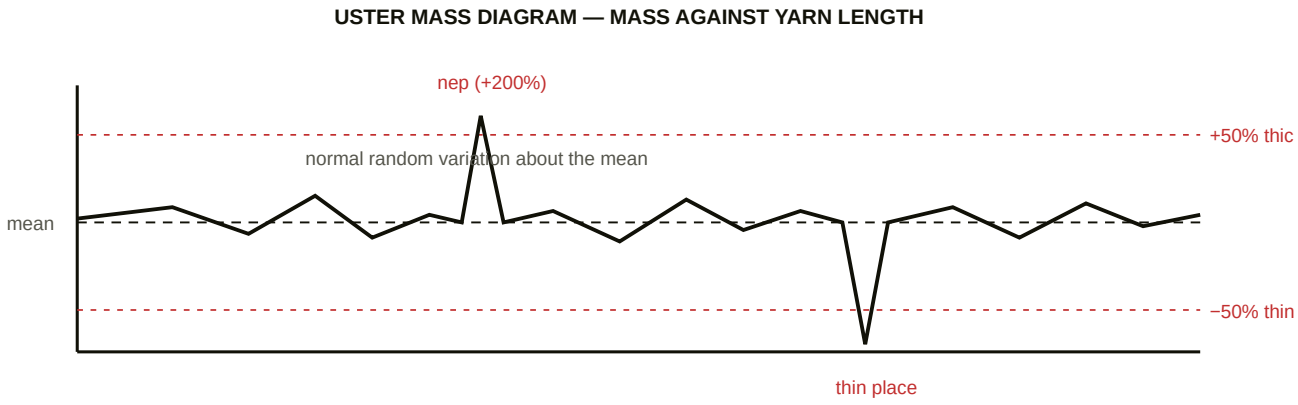


Figure 1 — The evenness tester's mass diagram: mass variation traced against yarn length, with the −50% and +50% imperfection thresholds, a nep spike reaching the +200% zone, and a thin place crossing the lower threshold.

2. Apparatus and procedure

- **Apparatus:** single-yarn strength tester (constant rate of extension, 500 mm gauge); lea strength tester with hooks; wrap reel and balance (count, from Experiment 2); capacitive evenness tester with diagram recorder and spectrogram; yarn fault classification data from the winder clearer or a fault board; conditioned yarn packages.
- **Single-end strength:** condition the package, draw off and discard the outer layers, and break at least 20 (preferably 50) specimens at spaced positions, at the standard gauge and rate. Reject jaw breaks — slippage or damage at the clamp measures the clamp, not the yarn. Record mean load, elongation and CV%.
- **Lea strength:** wind leas exactly as for count, place each lea evenly over the two hooks so all 80 strands share the load, extend to rupture and read the load in pounds. Average at least five leas. Multiply the mean load by the measured count for the C.S.P.
- **Evenness:** thread the yarn through the capacitive slot at the standard speed for its count, test a full package length (400 m is a common laboratory run), and record U%, CV_m, the three imperfection counts per km, the diagram and the spectrogram. Note any periodic peak and its wavelength before discarding the printout.
- **Fault analysis:** collect the clearer/classing data for a defined length of yarn (report per 100 km), distribute the faults over the Classimat length–size grid, and state which class dominates — small short faults, long thick faults or long thin faults.

3. Worked calculation

The yarn throughout is the 30 Ne of Experiment 2.

$$\text{Linear density: tex} = 590.5 / 30 \approx 19.68 \text{ tex}$$

Single-end strength. Fifty breaks give a mean breaking load of **245 cN**.

$$\text{Tenacity} = 245 / 19.68 \approx 12.45 \text{ cN/tex}$$

A tenacity of about 12.5 cN/tex is a respectable working value for a carded 30 Ne cotton yarn — sensibly below the fibre tenacity of Experiment 2 (≈26 g/tex ≈ 25.5 cN/tex), because in a yarn the fibres are helical, unequally loaded, and the break finds the weakest cross-section: roughly half the fibre strength is realised in the yarn, which is the normal spinning efficiency for cotton.

Lea strength and C.S.P. Mean lea breaking load = **78 lb** at count 30 Ne.

$$\text{C.S.P.} = 78 \times 30 = 2{,}340$$

A C.S.P. of 2,340 falls in the good band (2,200–2,500) for carded cotton yarn, in agreement with the single-end result: this is a sound yarn for its count, fit for weaving warp at moderate loom speeds or quality knitting.

Irregularity and imperfections. Evenness run on the same yarn:

QUANTITY	RESULT	NOTE
U%	11.8%	Good for carded 30 Ne.
CV_m	≈ 14.9%	Check: 1.25 × 11.8 = 14.75 ≈ 14.9 ✓ — the two statistics agree, as a near-normal mass distribution requires.
Thin places (−50%)	85 /km	Low — drafting is well controlled.
Thick places (+50%)	210 /km	Moderate; the largest IPI component.
Neps (+200%)	145 /km	Consistent with the 42 neps/g remaining in sliver (Experiment 4) after drafting.
Total imperfections (IPI)	85 + 210 + 145 = 440 /km	The single figure quoted when comparing yarns of the same count.

Fault classification (outline). Classing the clearer's cuts for this yarn over 100 km shows the expected pattern for carded cotton: the great majority of faults fall in the small short classes (A and B — short, moderately thick places from preparation), a small number in the larger short classes (C, D), and only occasional long faults (E–G) traced to piecings and a lapped roller, with long thin faults (H, I) rare. The

winding clearing limit is set to remove the objectionable classes economically: every cut costs a knot, so the limit is placed where the marginal fault removed is worth the marginal knot added.

> **CONSISTENCY OF THE WHOLE SHEET**
>
> Count 30 Ne → tex 19.68 → tenacity 12.45 cN/tex; lea 78 lb → C.S.P. 2,340; U% 11.8% → CV_m ≈ 14.9% by the 1.25 rule; IPI sums to 440/km. Four independent instruments tell one story — a good carded yarn — and each number carries its own internal cross-check.

4. Precautions

- Condition yarn before every test on this sheet: strength, count and evenness all move with regain, and unconditioned comparisons between packages are comparisons of weather.
- Reject jaw breaks in single-end testing and re-test; including them biases the mean downward and inflates the CV, punishing the yarn for the operator's clamping.
- On the lea tester, spread the strands evenly over both hooks and load at the standard rate; a bunched lea fails strand by strand at a fraction of its true load.
- Use the measured count — not the nominal count on the package ticket — in both the tex conversion and the C.S.P. A one-count error moves the C.S.P. by a full class boundary.
- Run the evenness tester at the speed and slot specified for the count, after warm-up and zero checks; a wrong slot compresses or rattles the yarn and manufactures irregularity that belongs to the instrument.
- Do not compare IPI or U% across very different counts: finer yarns are intrinsically more irregular because fewer fibres fill their cross-section. Compare within count, against the mill's control values and published statistics for that count.

Practice (viva preparation)

1. A 30 Ne yarn has a mean single-end breaking load of 245 cN. Find its tenacity in cN/tex.
2. A lea breaks at 78 lb and the count is 30 Ne. Find the C.S.P. and judge the yarn.
3. Why is the lea load per strand lower than 80 times the mean single-end load?
4. A yarn shows U% = 11.8%. Predict its CV_m and state the assumption behind the prediction.
5. Add up and interpret the imperfection counts: thin 85, thick 210, neps 145 per km.
6. A spectrogram shows one sharp peak at a fixed wavelength. What does it mean and how is the guilty part found?
7. Why is the winding clearer not simply set to cut out every fault, however small?

Answer key

1. tex = 590.5/30 ≈ 19.68 tex. Tenacity = 245/19.68 ≈ 12.45 cN/tex.
2. C.S.P. = 78 × 30 = 2,340 — in the good band (about 2,200–2,500) for a carded cotton yarn of this count.
3. The 80 strands never share the load perfectly equally — slight differences in strand length and tension load some strands more — and the specimen breaks progressively: the weakest, most-loaded strands fail first and throw their load onto the survivors. A bundle of strands is therefore weaker than the sum of its parts tested one by one; the lea value is an average behaviour figure, steadier but systematically lower per strand.
4. CV_m ≈ 1.25 × 11.8 ≈ 14.8% (the measured 14.9% agrees). The factor 1.25 is the ratio of standard deviation to mean deviation for a normal (Gaussian) distribution of mass along the yarn; strong disagreement between the two statistics warns of non-random, periodic content in the signal.
5. Total IPI = 85 + 210 + 145 = 440 imperfections per km. Thick places dominate, thin places are low (drafting is healthy), and the nep figure connects back to the sliver nep count of Experiment 4 — a coherent, good-quality carded yarn profile.
6. A single fixed wavelength means a periodic mass variation generated by one rotating machine element. The element is identified from the wavelength: at the delivery in question, wavelength = π × roller diameter for an eccentric or damaged roller (allowing for any draft between that roller and delivery), or from gear-tooth counts for a gearing fault. Replacing or regrinding that element removes the peak.
7. Every cut replaces a fault with a knot or splice, which is itself a (smaller) fault, a potential end-break in weaving, and lost production time. Clearing is therefore an economic balance: the limit is set to remove faults that would cause claims, breaks or visible fabric defects, and to leave sub-objectionable variation in place. Classimat analysis exists precisely to place that line from evidence.
